Reti Logiche = Reti comb. + Reti Sequenziali (Flip-Flop) → Registri, Contatori

Analisi, Sintesi

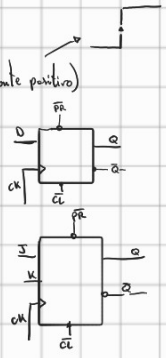
Registro permette di memorizzare + bit

Contatori contano i numeri naturali

Reti Sequenziali

F-F per (agisce ad ogni fronte positivo)

Usiamo solo i JK ed i D



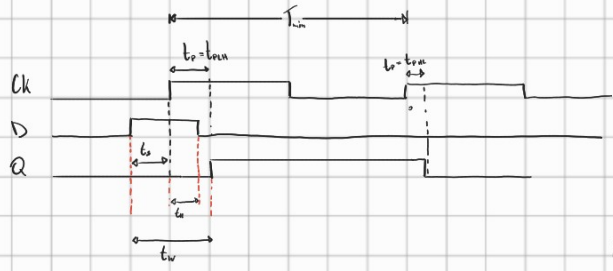
D	Q
0	0
1	1

J	K	Q_t
0	0	Q_{t-1}
0	1	0
1	0	1
1	1	$\overline{Q_{t-1}}$

(PRESET)
$\overline{PR}$ → forza il F-F allo stato "1" quando attivato (attivo basso)

(CLEAR)
$\overline{CL}$ → forza il F-F allo stato "0" quando attivato (attivo basso)

Temporizzazioni



T_{min} = frequenza massima del clock

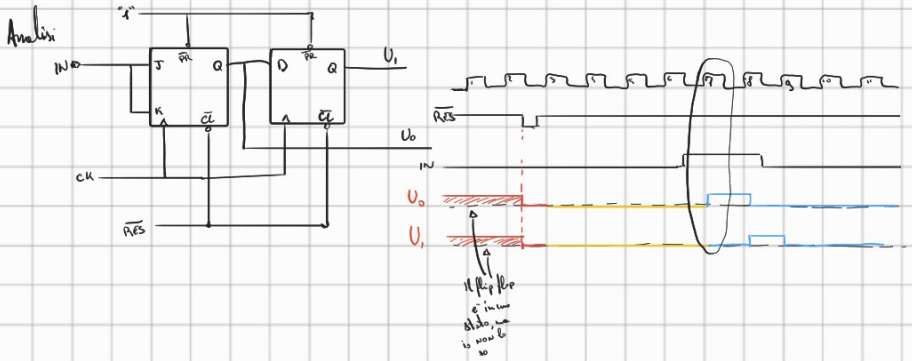
t_p = tempo di propagazione (max) → t_{PHL}, t_{PLH}

t_s = tempo di SET-UP (minimo) (min.)

t_H = tempo di HOLD (mantenimento) (min)

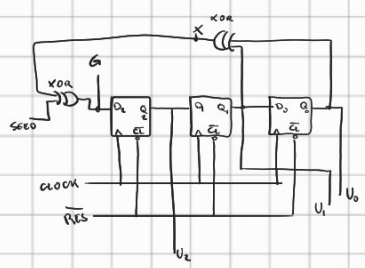
t_W = tempo di WIDTH (ampiezza dell'impulso) (min)

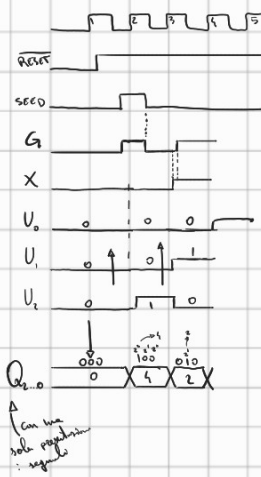
Analisi



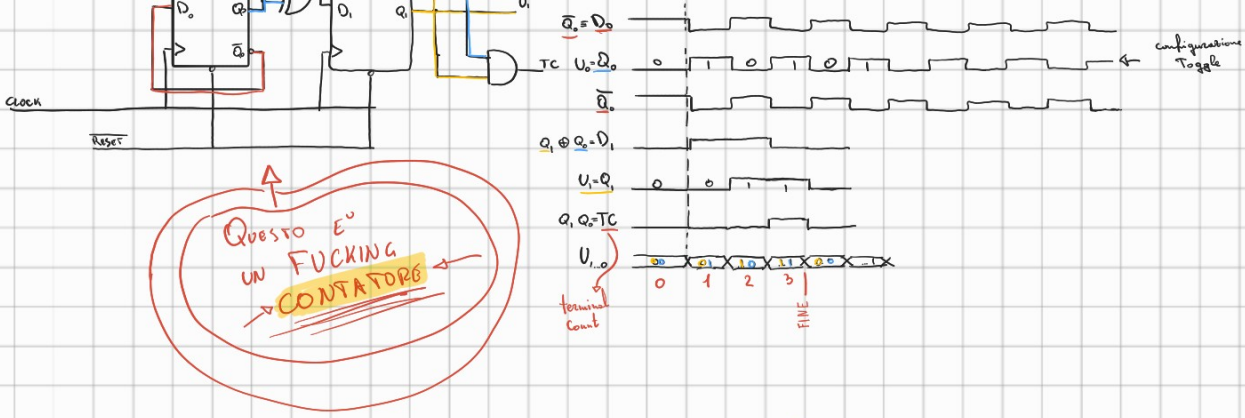
Analisi combinatoria-sequenziale

UNLIKE





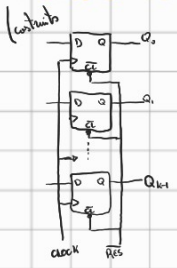




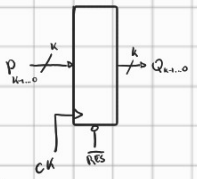
Blocchi costitutivi delle Reti Sequenziali:
- Registri
- Contatori

Registri:

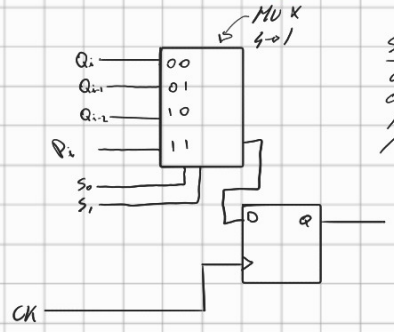
· Parallel Input Parallel Output (PIPO)



Registro PIPO a K bit

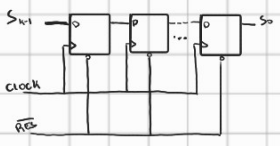


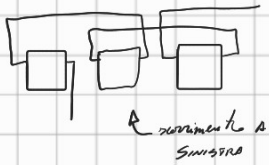
REGISTRO UNIVERSALE



S_0	S_1	Reg
0	0	Memoria
0	1	Scorrimento a DX
1	0	Scorrimento a SX
1	1	Caricamento Parallelo

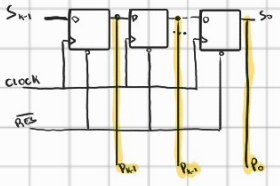
· Serial Input Serial Output (SISO) → Registro a scorrimento (a Destra)





Registro a scorrimento a SINISTRA

· Serial Input Parallel Output (SIPO)



· Parallel Input Serial Output (PISO)

